



Notifications

Notifications

- Android Notification provides short, timely information about the action happened in the application, even it is not running
- This application could be running in the background but not in use by the user.
- The purpose of a notification is to notify the user about a process that was initiated in the application either by the user or the system.
- The notification displays the icon, title and some amount of the content text.

Steps for notifications

Step 1 - Create Notification Builder

- As a first step is to create a notification builder using `NotificationCompat.Builder.build()`.
- You will use Notification Builder to set various Notification properties like its small and large icons, title, priority etc.

```
NotificationCompat.Builder mBuilder = new NotificationCompat.Builder(this)
```

Steps for notifications

Step 2 - Setting Notification Properties

- Once you have Builder object, you can set its Notification properties using Builder object as per your requirement.
- But this is mandatory to set at least following –
- A small icon, set by `setSmallIcon()`
- A title, set by `setContentTitle()`
- Detail text, set by `setContentText()`

```
mBuilder.setSmallIcon(R.drawable.notification_icon);
```

```
mBuilder.setContentTitle("Notification Alert, Click Me!");
```

```
mBuilder.setContentText("Hi, This is Android Notification Detail!");
```

Steps for notifications

Step 3 - Issue the notification

- you pass the Notification object to the system by calling **NotificationManager.notify()** to send your notification.
- Make sure you call **NotificationCompat.Builder.build()** method on builder object before notifying it.
- This method combines all of the options that have been set and return a new Notification object.

```
NotificationManager mNotificationManager =  
(NotificationManager) getSystemService(Context.NOTIFICATION_SERVICE);
```

```
// notificationID allows you to update the notification later on.  
mNotificationManager.notify(notificationID, mBuilder.build());
```

Steps for notifications

Step 4 – Create Notification Channel

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    tools:context=".MainActivity">
```

```
    <Button
```

```
        android:id="@+id/button"
```

```
        android:layout_width="wrap_content"
```

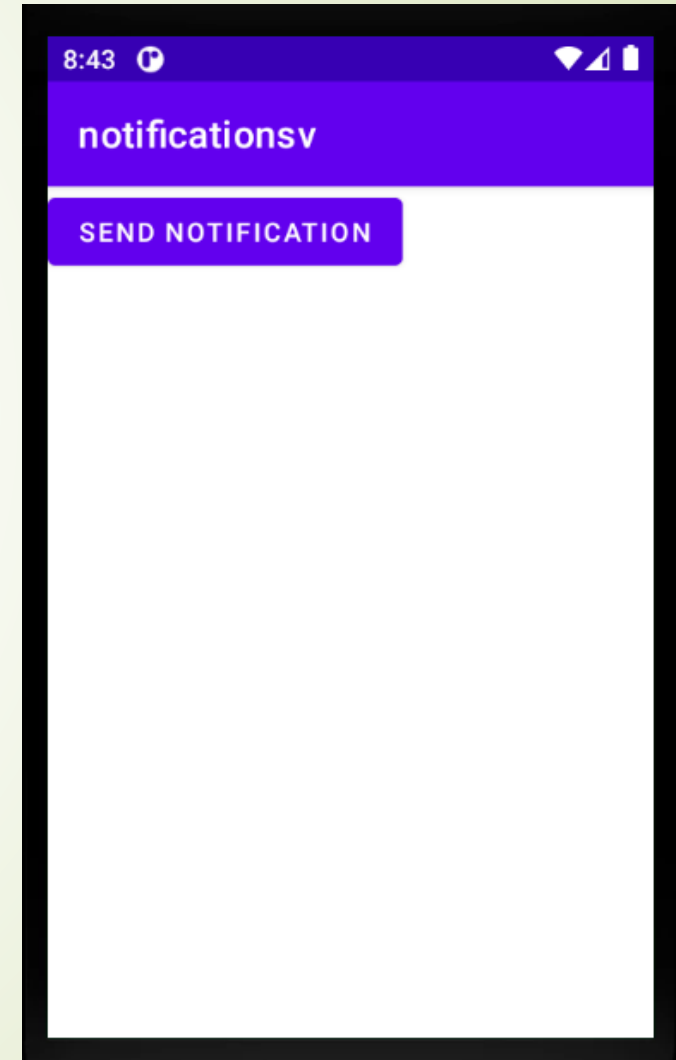
```
        android:layout_height="wrap_content"
```

```
        android:text="Send Notification"
```

```
        tools:layout_editor_absoluteX="180dp"
```

```
        tools:layout_editor_absoluteY="228dp" />
```

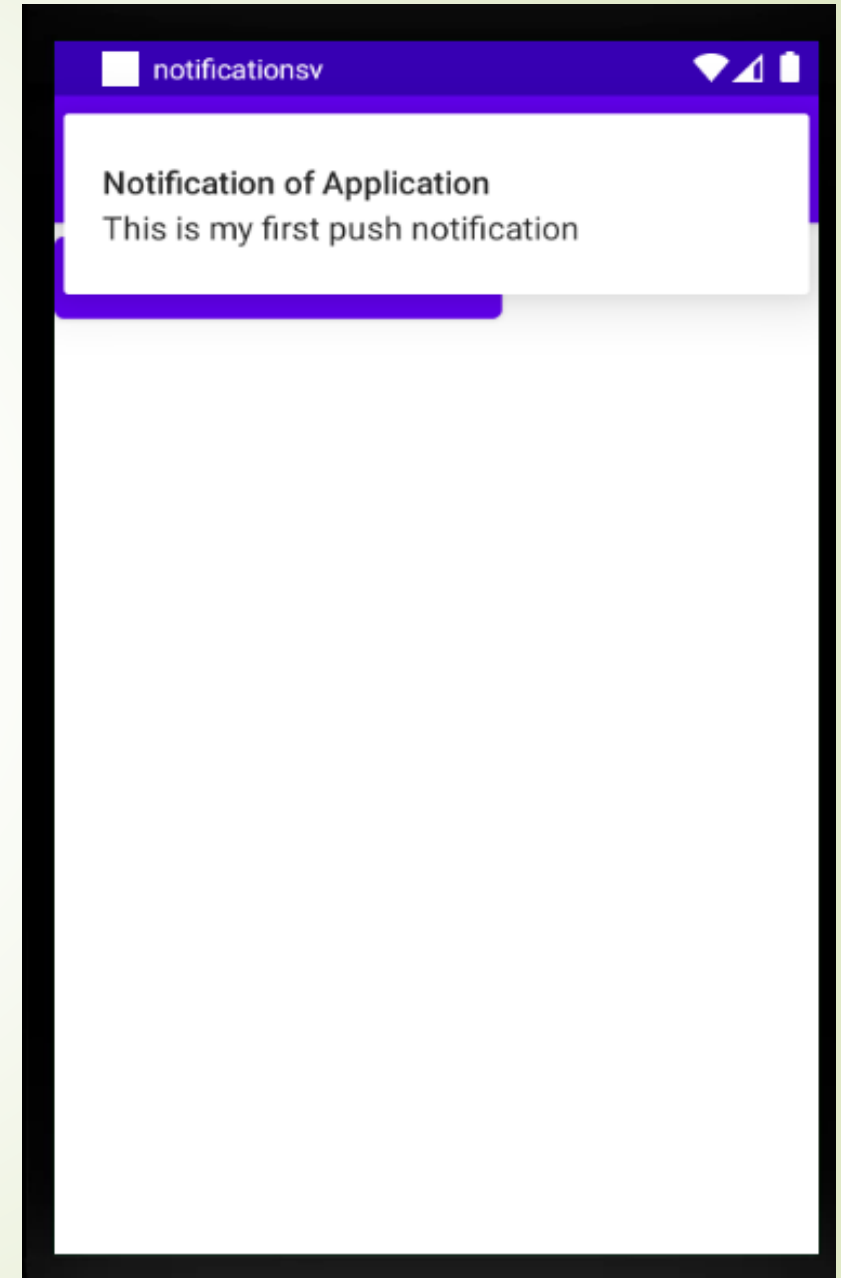
```
</RelativeLayout>
```



MainActivity.java

```
package com.example.notificationfinal;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.Context;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        String channel_id = "01" ;
        Button b = findViewById(R.id.button) ;
    }
}
```





```
@Override
public void onClick (View v) {
```

```
NotificationCompat.Builder mBuilder = new
```

```
NotificationCompat.Builder(MainActivity.this, "default" )
```

```
.setSmallIcon(R.drawable.logo)
```

```
.setContentTitle( "Notification of Application" )
```

```
.setContentText( "This is my first push notification" );
```

```
NotificationManager nm = (NotificationManager) getSystemService(Context.  
NOTIFICATION_SERVICE);
```

// checking if android version is greater than oreo(API 26) or not

if (android.os.Build.VERSION.SDK_INT >=

android.os.Build.VERSION_CODES.O)

{

NotificationChannel nc = new

NotificationChannel(channel_id , "CHANNEL_01" ,

NotificationManager.IMPORTANCE_HIGH) ;

mBuilder.setChannelId(channel_id) ;

nm.createNotificationChannel(nc) ;

}

nm.notify((int) System. currentTimeMillis (), mBuilder.build()) ;

}

}); } }